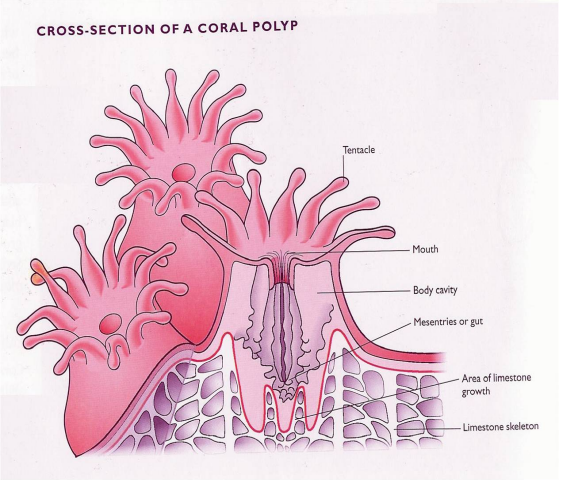
PART 2 – CORAL REEF ECOLOGY

- Coral reefs are rocky mounds and/or ridges formed in the sea by marine organisms through the accumulation and deposition of limestone (calcium carbonate).
- The reef framework provides the structural foundation of a unique and rich marine ecosystem. A single reef may cover over100 sq km.
- The living reef forms the top layer of the reef adding new limestone to the reef.
- These specialized habitats provide shelter, food, and breeding sites for numerous plants and animals and form a breakwater for the adjacent coast, providing natural storm protection



Great Barrier Reef

Stony corals are the major reef architects. These small marine animals, (individual organisms are called **polyps)**, produce a hard skeleton made of calcium carbonate, which they extract from the seawater and combine with CO^2 for limestone.



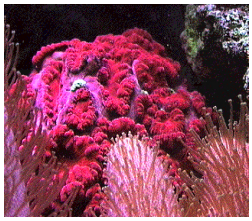
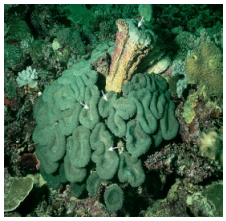
Coral Types

- Hard (Stony, scleractinian, "true") corals build the reef by extracting calcium carbonate from the ocean water and they create a diverse 3-D habitat for many other organisms
- Coral are actually a special group of cnidarians
- ***Hermatypic Corals*** - Corals that form large colonies called reefs and have a symbiotic relationship with the dinoflagellate Zooxanthellae
- ***Ahermatypic Corals*** - Corals that are solitary or form small colonies- they often lack the symbiotic relationship with Zooxanthellae and do not help build reefs
- All the ***different colors and shapes*** made up of thousands of individual polyps, each secreting its own small cup of coral limestone, which provide the building blocks for reef construction.



Mutualism between the Coral Polyp and Zooxanthellae – group of dinoflagellates

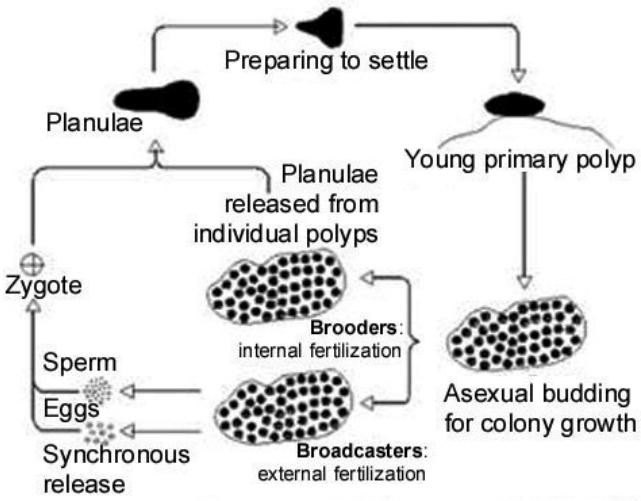
- **Coral Polyp** provides a home for the zooxanthellae, it provides nitrates and phosphates, and it gives off CO_2 – 90% of the coral's nutrients
- **Zooxanthellae** carries out photosynthesis and make oxygen and food for the polyp through photosynthesis, gain nutrients from the corals nitrogen and phosphorus wastes, and provide for most of the colors for the coral in the reef making them look like underwater gardens

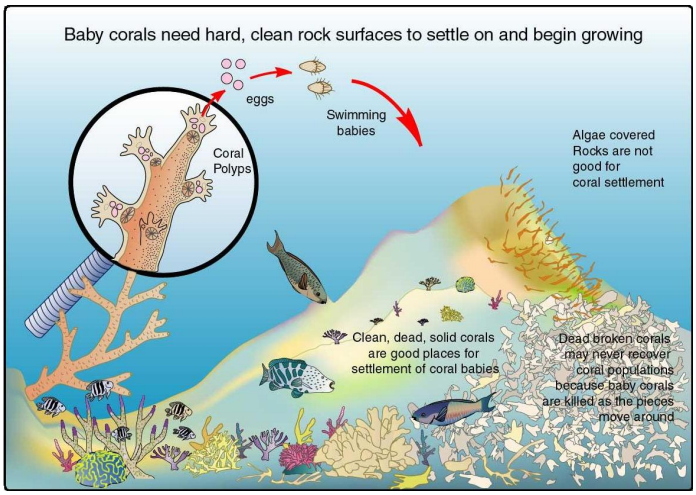

Coral Life Cycle and Reproduction

- First stage of the coral's life cycle is **planula larvae**, which allows it to be free swimming.
- Second stage of its life is **polyp** which is when the coral is stuck to a rock.
- In the polyp stage, it is able to reproduce,
  - either **asexual** - involves the splitting of a coral (called fission) or sprouting another coral from itself (called budding).
  - **sexually** (with another polyp)
  - involves a cycle of:

SPAWNING >> FERTILIZING >> PLANULAE LARVAE SETTLEMENT >> CLONING.



Planulae Settlement



Other Reef Building Organisms

- Fire corals
- Blue and Organ Pipe corals
- Coralline algae form cementing crusts that act as 'mortar' for the coral 'blocks'
- Aggregations of the tropical reef worm (*Phragmatopoma lapidosa*) construct low reefs called **Worm Reefs** of tubes consisting of sand grains cemented together by protein. The reefs expand as worm larva settle on existing tube masses. The reef growth is controlled by waves bringing planktonic food and sand to the worms and are found from Cape Canaveral to Key Biscayne

Requirements for Reef Formation

- A solid structure for the base with a hard substrate for attachment
- Warm and predictable water temperatures > 20°C (68°F) and oceanic salinities
- High Light Levels
- Clear waters with high water transparency
- Low nutrient waters - low in phosphate and nitrogen nutrients
- Good water circulation with moderate wave action to disperse wastes and bring oxygen and plankton to the reef

Major Areas of Coral Reef Development

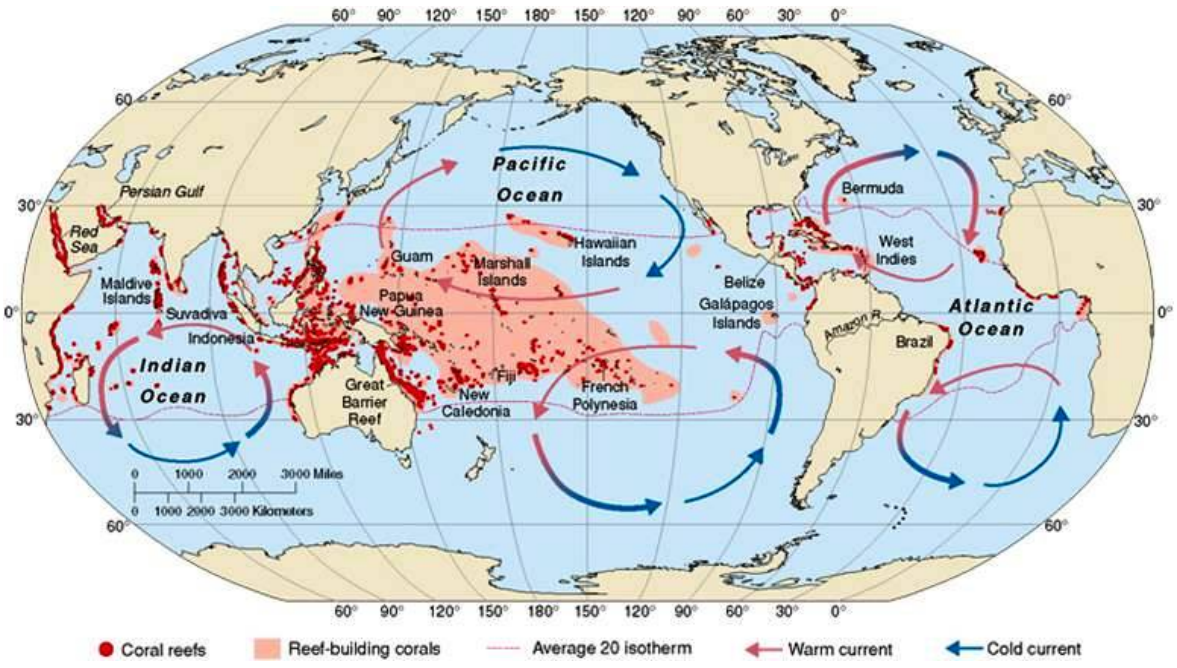
- Shallow submarine platforms in the tropics
- The best conditions for coral reef development are concentrated towards the western ends of the three major ocean basins (Atlantic, Pacific and Indian) so this is where most of the world's coral reefs are to be found

Three Major Regions with great diversity of organisms

- ***Indo Pacific Region*** **–** is the largest of the three major regions in center of map below
- ***Red Sea*** **–** out pocket of the Indian Ocean in far west portion of the ocean basin
- ***Greater Caribbean Region*** of the western Atlantic

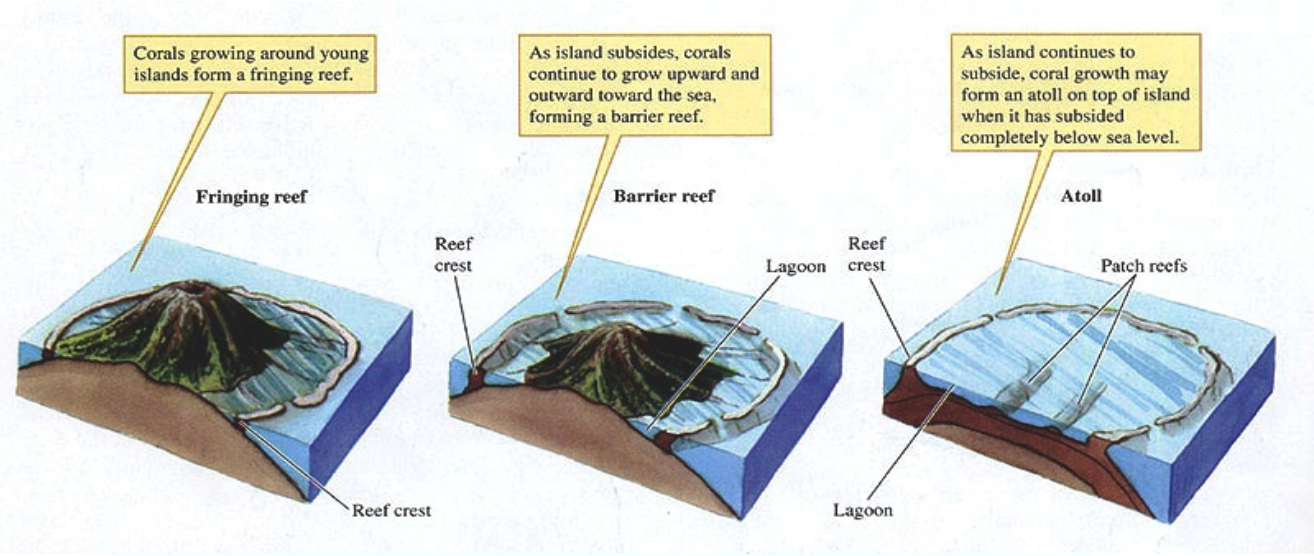
Minor Regions of smaller fragmented areas of coral reef development

- ***Eastern Pacific, off Western Australia, Southern Japan in the Pacific Ocean***
- ***Tropical eastern Atlantic, East coast of southern Brazil, Island of Bermuda in western Atlantic***
- These areas are at the extreme margins of the ecological tolerances of hard corals, where environmental conditions are only minimally capable of sustaining only a fraction of the hard coral species found in the two main regions of reef development



Stages in Coral Reef Development – Three Basic Types of Coral Reefs

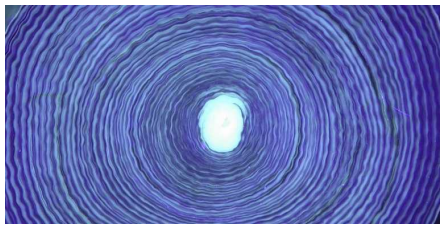
They begin with a brand new tropical island produced by an oceanic hot spot or at a plate boundary and gradually change through thousands of years from a ***fringing reef***, to a ***barrier reef***, to ***an atoll***, and finally to an extinct reef as a ***seamount*** (an underwater mountain) or ***guyot*** (underwater mountain or seamount with a flat top) A ***patch reef*** is an isolated coral growth forming a small platform in a lagoon, barrier reef, or atoll





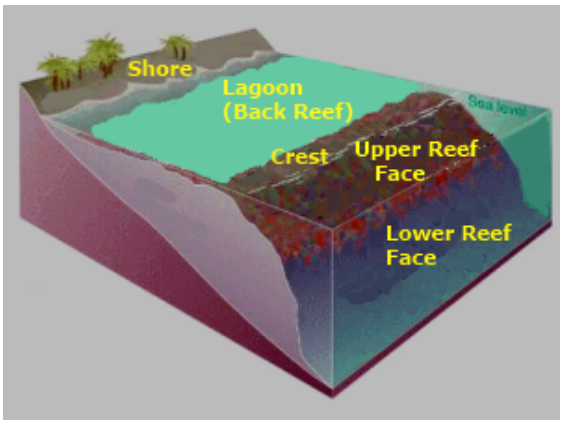
Atoll with Patch Reefs **Fringing Reef** **Barrier Reef**

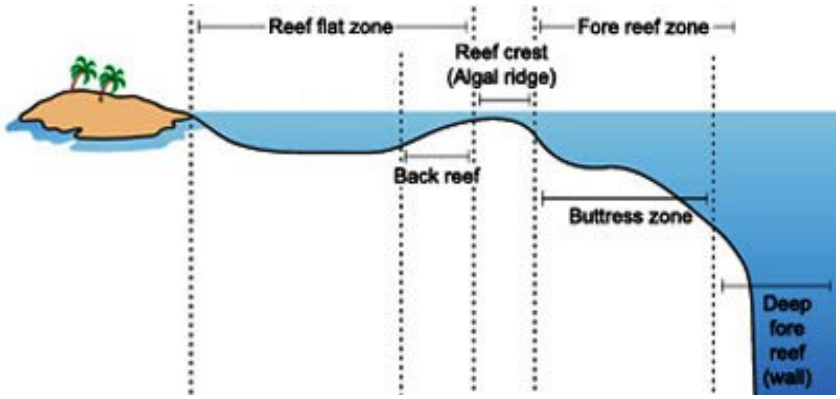
- The largest coral reef, the Great Barrier Reef, is 1,250 miles long
- Large reefs grow at the rate of 1 to 2 cm per year
- It's estimated that some of the largest reefs took as long as 30 million years to form
- Scientists are aging reefs by counting the coral growth layers (like tree rings) **Coral growth patterns:** The growth rate and density of coral skeletons also vary with temperature and other environmental conditions, so their growth patterns can be analyzed much like tree rings. The chemical composition of the coral in a particular ring provides information about sea surface temperatures, salinity, runoff, and upwelling.



Zones of A Coral Biome

- Coral reefs have the greatest diversity of marine life of any ocean biome and are often called the rainforests of the ocean.
- A relatively small biome, but around 25% of the known marine species live in coral reefs.
- The largest coral reef, the Great Barrier Reef, has1500 species of fish and 400 species of coral

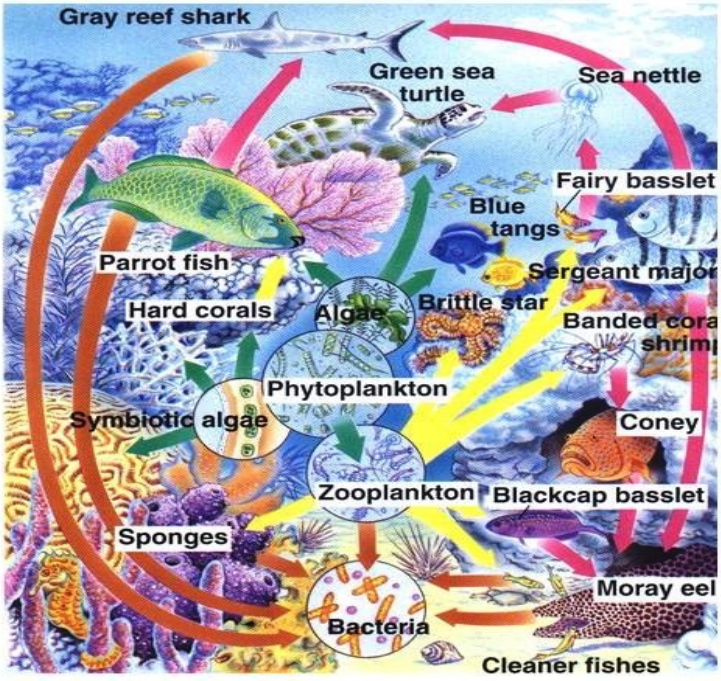




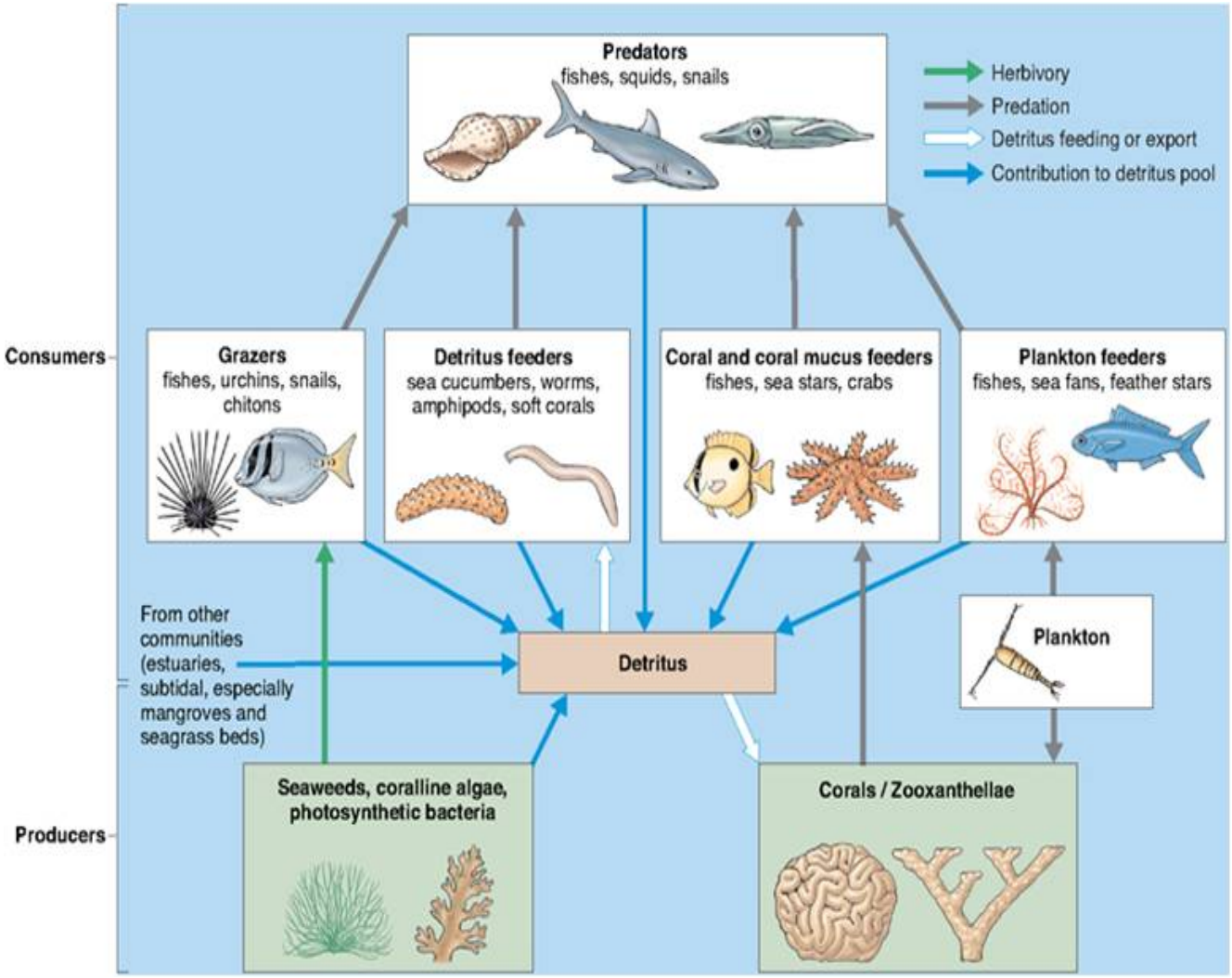
- **Shore or inner reef zone** - area is between the crest and the shoreline. Depending on the shape of the reef, this area can be full of life including fishes, sea cucumbers, starfish, and anemones.
- **Crest reef zone** - highest point of the reef and where the waves break over the reef.
- **Fore or outer reef zone** - As the reef wall falls off, the waters get calmer. Around 30 feet deep, will be the most populated part of the reef along with lots of different types of coral species.

Coral Reef Communities -Coral reefs are inhabited by thousands of species including:

- Algae
- Sponges
- Soft corals
- Sea slugs
- Urchins and star fish
- Worms
- Crabs and lobster
- Snails
- Clams, scallops, and barnacles
- Fish
- Sea turtles
- Sharks and rays



Coral Reef Food Web



Coral Reef Fish Communities

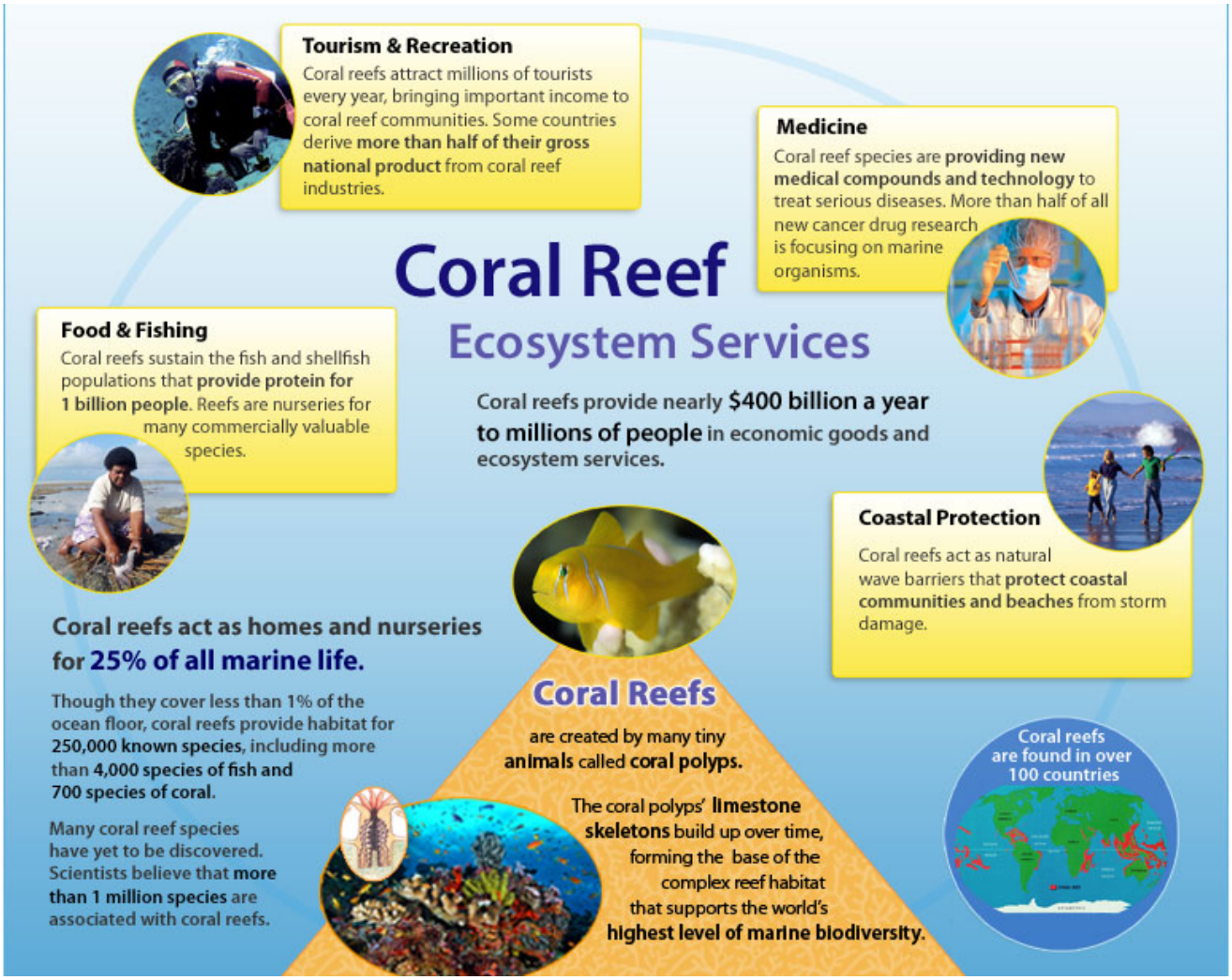
- Hundreds of species can exist in a small area of a healthy reef, many of them hidden or well camouflaged.
- Reef fish have developed many ingenious specializations adapted to survival on the reefs.
- They provide a home for 25 percent of all marine fish species
- Loss and degradation of coral reef habitat, increasing pollution, and overfishing including the use of destructive fishing practices, are threatening the survival of coral reefs and the associated reef fish.



the

Importance of Coral Reefs Includes

- Fishery and nursery areas (food) are important sources of protein for people locally and globally
- Tourism and recreation bring coral reef experiences to people from far away, and is an important part of the local economy
- Potential medicines such as medicines for cancer are very promising areas of current research
- Coastal protection - protection of coastlines from erosion and storm damage



Coral Reef Health Indicators

- Marine Apex Predators
- Biomass
- Average Catch Length
- Coral Cover
- Indicator Organisms

Coral Reef Indicator Organisms

- Regions of the world have established indicator organisms for coral reefs.

- In **1996 Reef Check** developed a list of world-wide species which is used my many counties as a basis for regional indicator lists – the list was chosen to help recognize overfishing, blast fishing, poison fishing, aquarium fish collection, nutrient pollution, and curio collection

- **Indicator Organisms:**

  Global

  Banded coral shrimp (***Stenopus hispidus***)
  Butterfly fish (***Chaetodon* spp.)**
  Crown of thorns starfish (***Acanthaster planci***)
  Fleshy algae (**Rhodophyta, Chlorophyta, Phaephyta**)
  Grouper >30 cm (Serranidae, Epinephelinae)
  Hard **/Stony** corals
  Lobsters (**Nephropidae**)
  Long-spined black sea urchins (Diadematideae/***Diadema* spp**.)
  Morey eel (*Muraenidae*)
  Parrotfish (>20 cm) (Scaridae)
  Pencil urchins (**Eucidairis spp**.)
  Snapper (*Lutjanidae*)
  Demosponges (**Demospongiae**)
  Sweetlips - (*Haemulidae*)
  Triton *(**Charoniidae**/***Charonia* spp**.)

  Indo-pacific region only

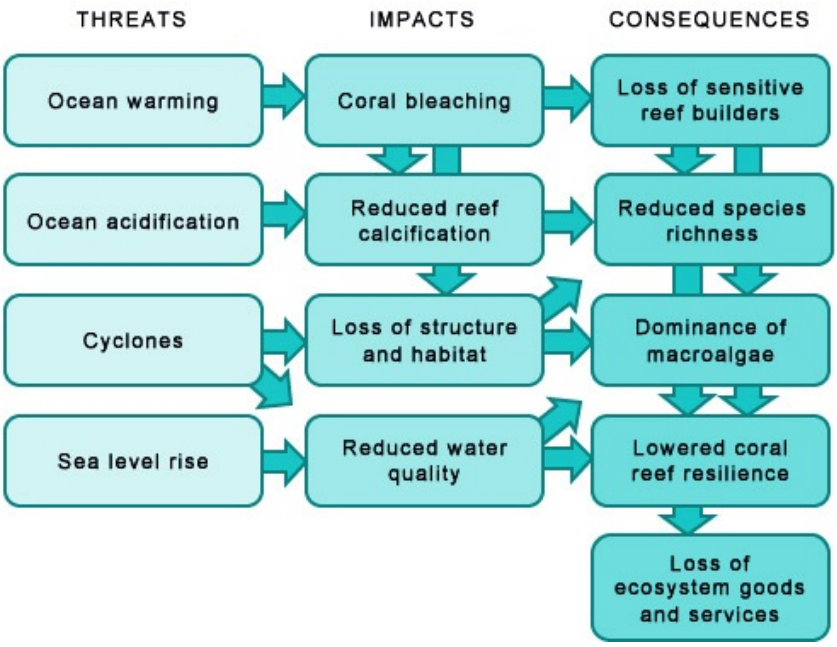
  Barramundi cod (***Cromileptes altiverlis***)
  Bumphead parrot (***Bolbometopon muricatum***)
  Giant clams (***Tridacna* spp.**)
  Humphead wrasse (***Cheilinus undulatus***)
  Sea Cucumber (***Thelenota ananas, Stichopus chloronotus***)

  Atlantic region only
  Gorgonia
  Flamingo Tongue Snail (***Cyphoma gibbosum***)
  Nassau grouper (***Epinephelus striatus***)

Coral Reef Threats

- Pollution in the form of chemical pollutants and excess nutrients
- Dredging and Sedimentation
- Coral diseases such as white band disease and Stony Coral Tissue Loss Disease (SCTLD)
- Climate change and ocean acidification: may lead to coral bleaching
- Overfishing disturbed natural food chains and food webs
- Coral Reef damage by cruise ship anchors and sewage, by tourists breaking off chunks of coral, and by commercial harvesting for sale to tourists. One study of a cruise ship anchor dropped in a coral reef for one day found an area about half the size of a football field completely destroyed, and half again as much covered by rubble that died later. It was estimated that coral recovery would take fifty years.

Threats to Coral Reefs, Their Impacts and Consequences





Healthy reef



Damaged Reef

Coral Reef Management

- Fisheries regulation
- Marine protected areas
- Coastal zoning
- The problem of ecosystem phase-shifts (how if corals die and area is taken over by algae, it achieves a new steady state and is very difficult for corals to re-colonize)

International Year of the Reef 2008



Phytoplankton
1. Diatom, *Chaetoceros sp.*
2. Foraminifera
3. Acantharian amoeba
4. Dinoflagellates

Zooplankton
5. Copepod
5.' Copepod nauplius larvae
6. Crab zoea larvae
7. Gastropod velliger larvae
8. Foraminifera
9. Radiolarian

Reef-building corals
10. Blue octocorals, *Heliopora coerula*
11. *Acropora sp.*
12. *Platygyra daedalea*
13. *Astreopora sp.* & *Pocillopora sp.*
14. *Pocillopora elegans*
15. *Lobophyllia corymbosa*
16. *Goniastrea sp.* & *Acropora sp.*

Macroalgae
18. Crustose coralline algae
19. *Dictyota sp.*
20. *Padina sp.*
21. *Halimeda sp.*

Fish
23. Pyramid butterflyfish
24. Blue sprat, "baitfish"
25. Yellowtail fusiliers
26. Surgeonfish
27. Scribbled rabbitfish
28. Reticulated & Threadfin butterflyfish
29. Steephead parrotfish
30. Slingjaw wrasse
31. Humphead Wrasse
32. Titan triggerfish
33. Scrawled filefish
34. Regal angelfish
35. Porcupine pufferfish
36. Yellowstripe goatfish
37. Yellowtail wrass
38. Redmouth grouper
39. Blackfin barracuda
40. Blacktip shark
41. Spotted eagle ray

Reptiles
42. Hawksbill sea turtle

Other invertebrates
17. Giant Clam, *Tridacna sp.*
43. Crab, *Trapezia sp.*
44. Snail, *Drupella sp.*
45. Crown of thorns seastar, *Acanthaster planci*
46. Cushion star, *Culcita novaguinea*
47. Rock-burrowing urchin, *Echinometra sp.*
48. Diademnid urchin, *Echinothrix sp.*
49. Crinoid, *Comanthus sp.*
50. Gorgonian seafan, *Arinella mollis*
51. "Featherduster" tubeworm, *Sabellid sp.*
52. Sponge, *Aplysilla sp.*
53. Lace coral, *Stylaster sp.*
54. Crab, *Carpilius maculatus*
55. Lobster, *Panulirus sp.*
56. Tiger Cowrie, *Cyprae tigris*
57. Triton trumpet, *Charonia tritonis*
58. Jellyfish, *Aurelia aurita*
59. Hermit crab
60. Reef Squid, *Sepioteuthis lessoniana*
61. Tun snail, *Tonna perdix*
62. Octopus, *Octopus cyanea*
63. Sea-cucumber, *Thelenota ananas*
64. Auger snail, *Terebra maculata*
65. Digested sediment from Peanut worm, *Echiuroid sp.*
66. Heart urchin
67. Mantis shrimp
68. Annelid worm, *Pherecarida sp.*



Coral Reef Food Web

Zooanthellae are tiny single-celled algae that live inside of corals and clams.

These algae provide ~ 70% of the food for these animals.

Zooanthellae form the base of the





Tiny polyp of a table coral with millions of zooanthellae living within the coral host tissue.